

The Vertical Evolution of the Yellow Dragon: Quantifying the AOD-to-Surface PM Discrepancy during the April 2025 Dust Event

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Disclaimer: This set of slides is taken from a group project and mainly presents the work on vertical evolution of dust storm that I have mainly contributed to.

Outline

- 1 Background
- 2 Simulation Setup
- 3 Vertical Transportation and Structure

Time-Height Diagrams

Vertical Profile of the Yellow Dragon at Peaks

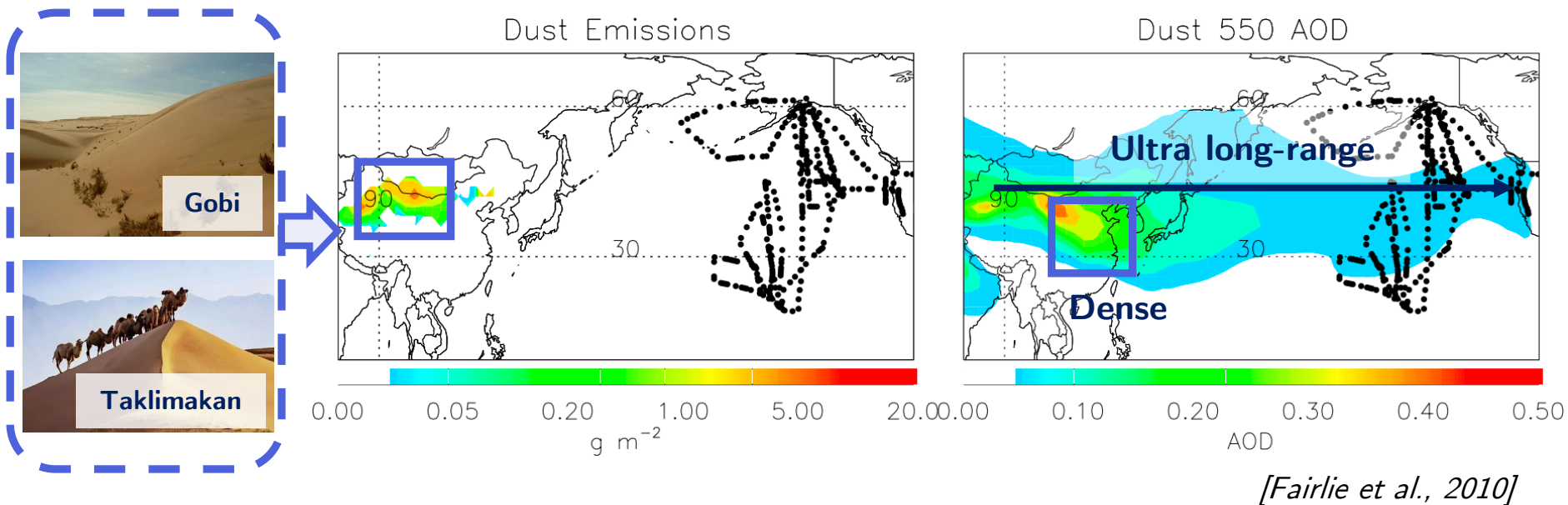
Size-Resolved & Latitudinal Dust Cross-Section Analysis

Column AOD and Surface Dust Relationship

Background

Mineral Dust in the Atmosphere

Mineral dust is the **largest single contributor** to atmospheric particulate matter. It has been extensively studied because of its **profound impacts on air quality, human health, climate, and marine ecosystems.**



- ▶ Dust from Asian dust events can be transported **over a long distance.**
- ▶ The downwind region of Asian dust includes many **major cities with dense populations.**

A Severe Dust Storm During April 2025 in China



GeneOnline News

<https://geneonline.news> › 最新文章 · 转为简体网页

This event, one of the *most severe* in recent years, provides an excellent opportunity to quantify the impact of dust.

2025年中國北方大規模沙塵暴侵襲影響波及多國

2025年10月7日 — 2025年4月的中國沙塵暴災難，是自然因素與人為因素共同作用的結果。氣候變遷加劇了乾旱，不合理的土地利用方式則擴大了沙源地。其影響不僅限於環境，更深入到 ...



Facebook · 香港地下天文台

1550+ 个回应 · 12个月前

[12/4/2025 23:30HKT] 【沙塵天氣正逐漸影響香港】 ...

香港方面，午夜前多區的可吸入懸浮粒子濃度正逐漸上升，預計4月13日本港空氣質素會顯著轉差，高危人士可能會出現較嚴重的健康反應，而公眾亦可能會出現 ...

Our group provides insights on this event, through analyzing both *vertical and surface transport distributions*, and assessing its *impacts on air quality in China (PM_{2.5} and PM₁₀)*.

Simulation Setup

Control Run Setup

- **geoschem_config.yml**: event window (1-22 April 2025), region (entire China).

```
start_date: [20250401, 000000]
end_date:   [20250423, 000000]
grid:
  resolution: 0.5x0.625
  number_of_levels: 47
  longitude:
    range: [ 65.0, 140.0]
    center_at_180: true
```

With **nested simulation**: covers dust **source regions** (northern China, Inner Mongolia) and **downwind areas** (eastern/southern China).

```
latitude:
  range: [ 8.0, 58.0]
  half_size_polar_boxes: true
nested_grid_simulation:
  activate: true
  buffer_zone_NSEW: [3, 3, 3, 3]
```

- **HISTORY.rc**

```
DustEvent.template:      '%y4%m2%d2_%h2%n2z.nc4',
DustEvent.frequency:     00000000 010000
DustEvent.duration:      00000001 000000
DustEvent.mode:          'time-averaged'
DustEvent.fields:
  'SpeciesConcVV_DST1    DST1      ',
  'SpeciesConcVV_DST2    DST2      ',
  'SpeciesConcVV_DST3    DST3      ',
  'SpeciesConcVV_DST4    DST4      ',
  'PM25                  PM25      ',
  'PM10                  PM10      ',
  'Met_PBLH              PBLH      ',
  'Met_U                 U         ',
  'Met_V                 V         ',
  'Met_T                 T         ',
  'Met_PMID              PMID      ',
::
```

Outputs:

3D dust mass concentration under the multi-bin dust scheme, surface PM2.5, PM10, PBL height, winds, temperature, pressure.

Control Run Setup

► HEMCO_Config.rc

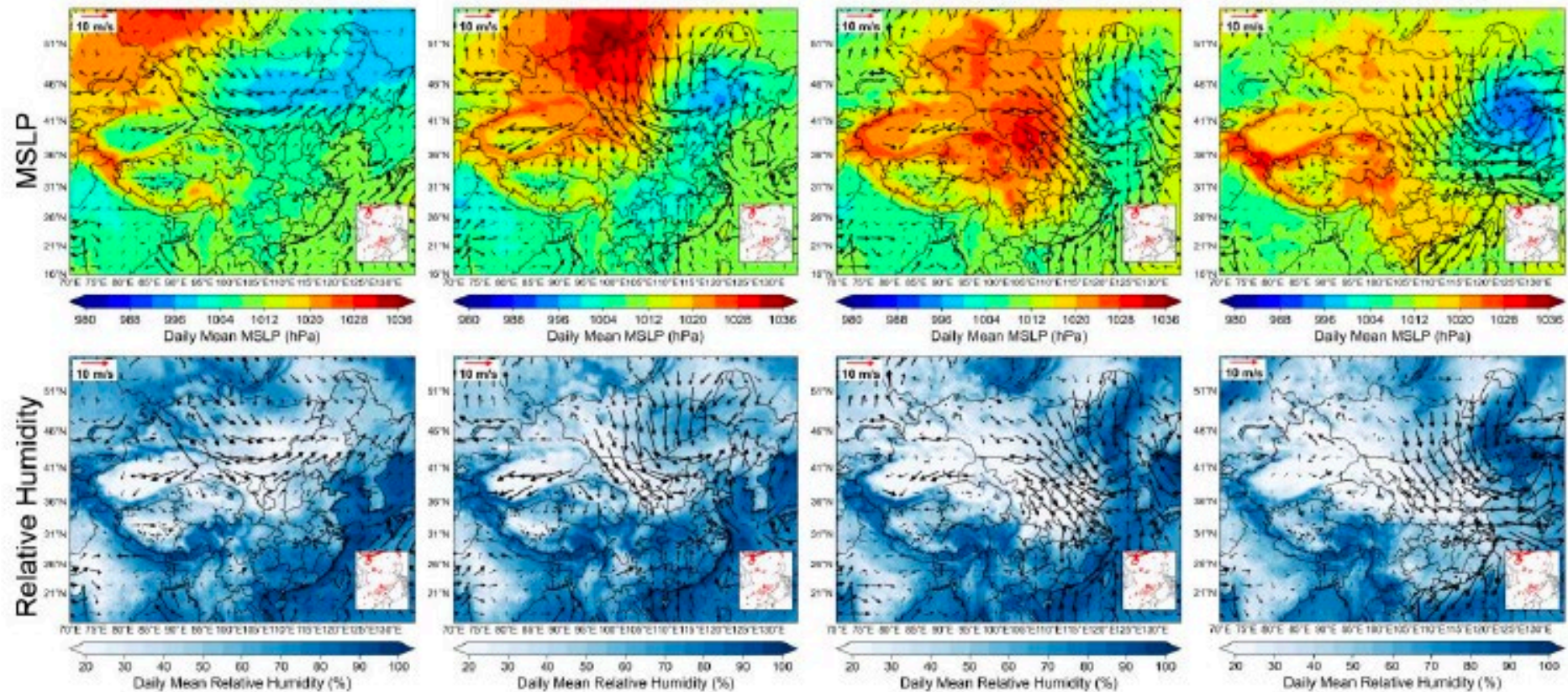
```
--> OFFLINE_DUST      :      true      # 1980-2019
--> OFFLINE_SEASALT    :      true      # 1980-2019
--> GFED4_CLIMATOLOGY  :      true      # If true, turn off GFED ext below
111    GFED            : off          ...
117    Volcano          : on          SO2
```

Dust emissions are read from **pre-computed** files (based on the selected meteorology MERRA-2).

- 3-month **spin-up** period for both **global** simulation ($4^{\circ}\times 5^{\circ}$) and **nested** simulation ($0.5^{\circ}\times 0.625^{\circ}$)

Results and Analysis

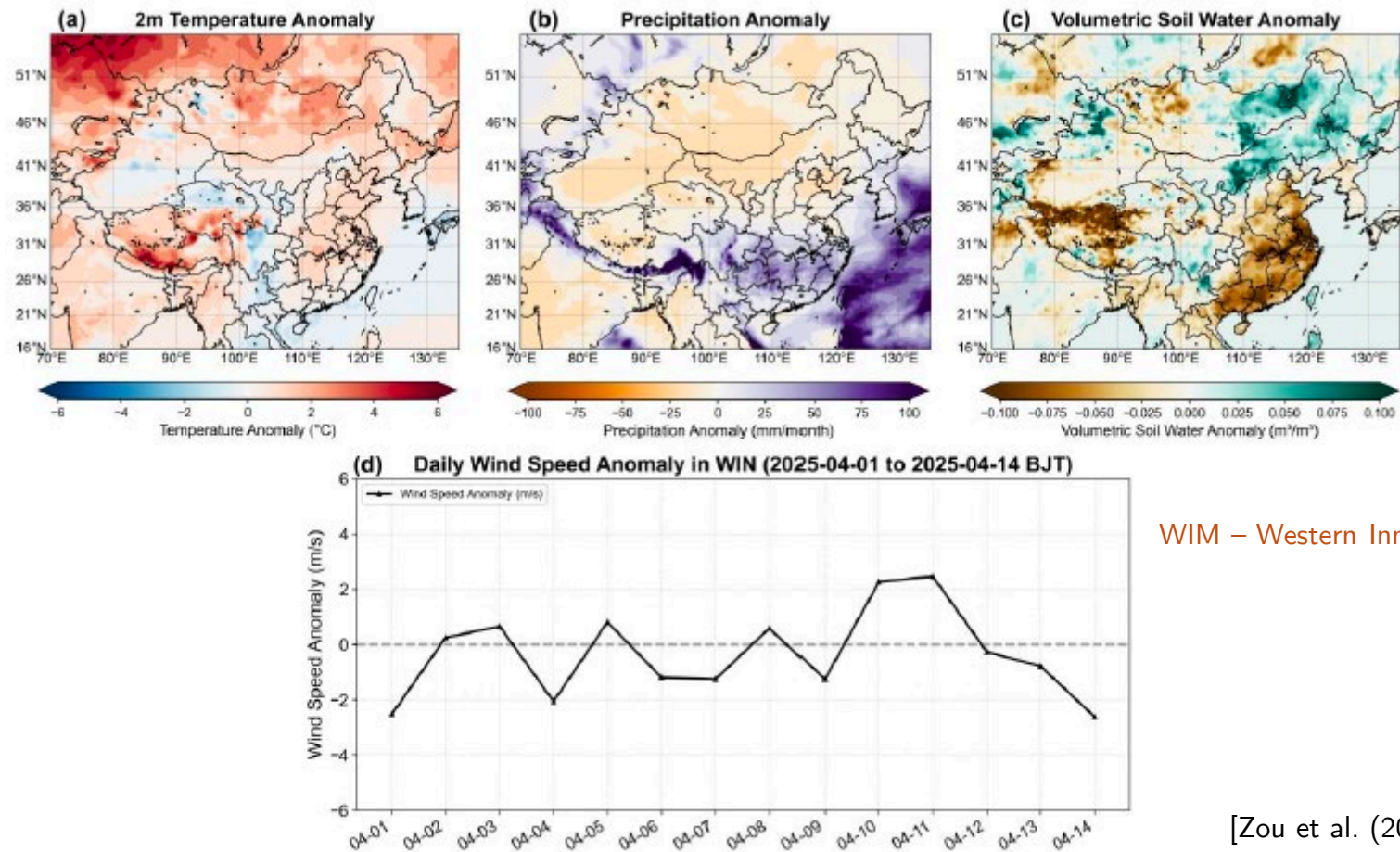
Synoptic Setup Behind the April 2025 Dust Event



[Zou et al. (2026 preprint)]

- **Siberian high + Mongolian cyclone** controlled the event.
- **Dry-cold advection and strong northerly winds** favored dust lifting and transport.
- Dust stayed behind the rainband, so **wet scavenging was weak**.

Why Was the 2025 Event So Strong?



- ▶ The source region was already **warm and dry before the event**
- ▶ Short-term **wind-speed anomalies** further strengthened the outbreak
- ▶ Surface preconditioning + strong winds made the 2025 event unusually severe

Vertical Transport – Research Question

Motivations:

- ▶ Validating model vertical profiles against event-specific observations (or satellite AOD) helps **improve future forecasts**.
- ▶ **Satellite AOD is often used as a proxy for air quality** in regions without ground monitors. However, satellite AOD alone cannot reliably estimate surface PM concentrations during dust events because the **vertical distribution of dust decouples the column from the surface**.

Research Question: How does the **vertical distribution** of natural dust aerosols **evolve during long-range transport**, and what are the **implications for interpreting satellite AOD vs. surface PM concentrations** during the April 2025 East Asia dust event?

- ▶ How does the **dust layer height** change as the plume moves from source (Inner Mongolia) to far downwind (Hainan)?
- ▶ What are the **implications for interpreting satellite AOD vs. surface PM concentrations** during the April 2025 East Asia dust event?
- ▶ Can we identify a **“transport efficiency” factor** (AOD per unit surface concentration) that varies with distance?

Transport Pathway Reconstruction

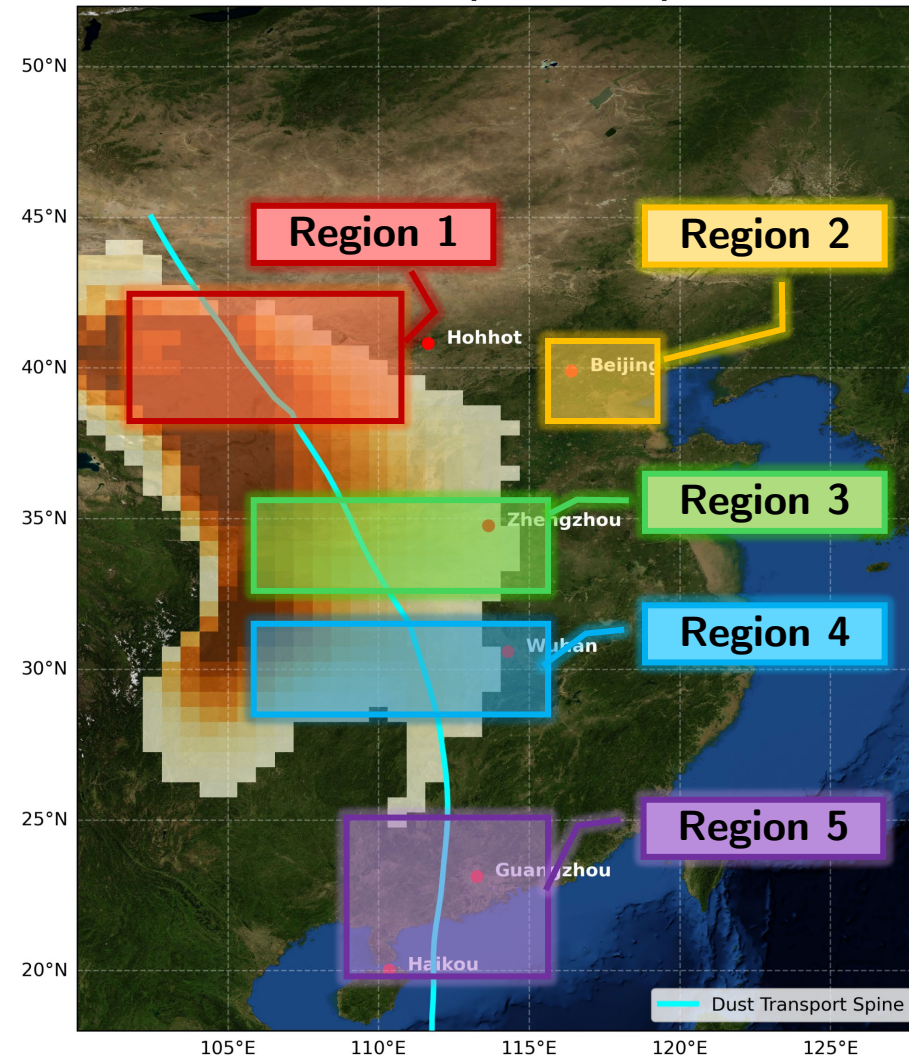
Reconstruction Attempt:

1. **Sum over time and the maximum over vertical levels**, the dust mass mixing ratio (all **DST*** tracers) for an accumulated footprint.
2. **Extracts a transport spine** using a latitude-dependent longitudinal search window (forcing a western anchor at high latitudes, allowing eastward drift southward).

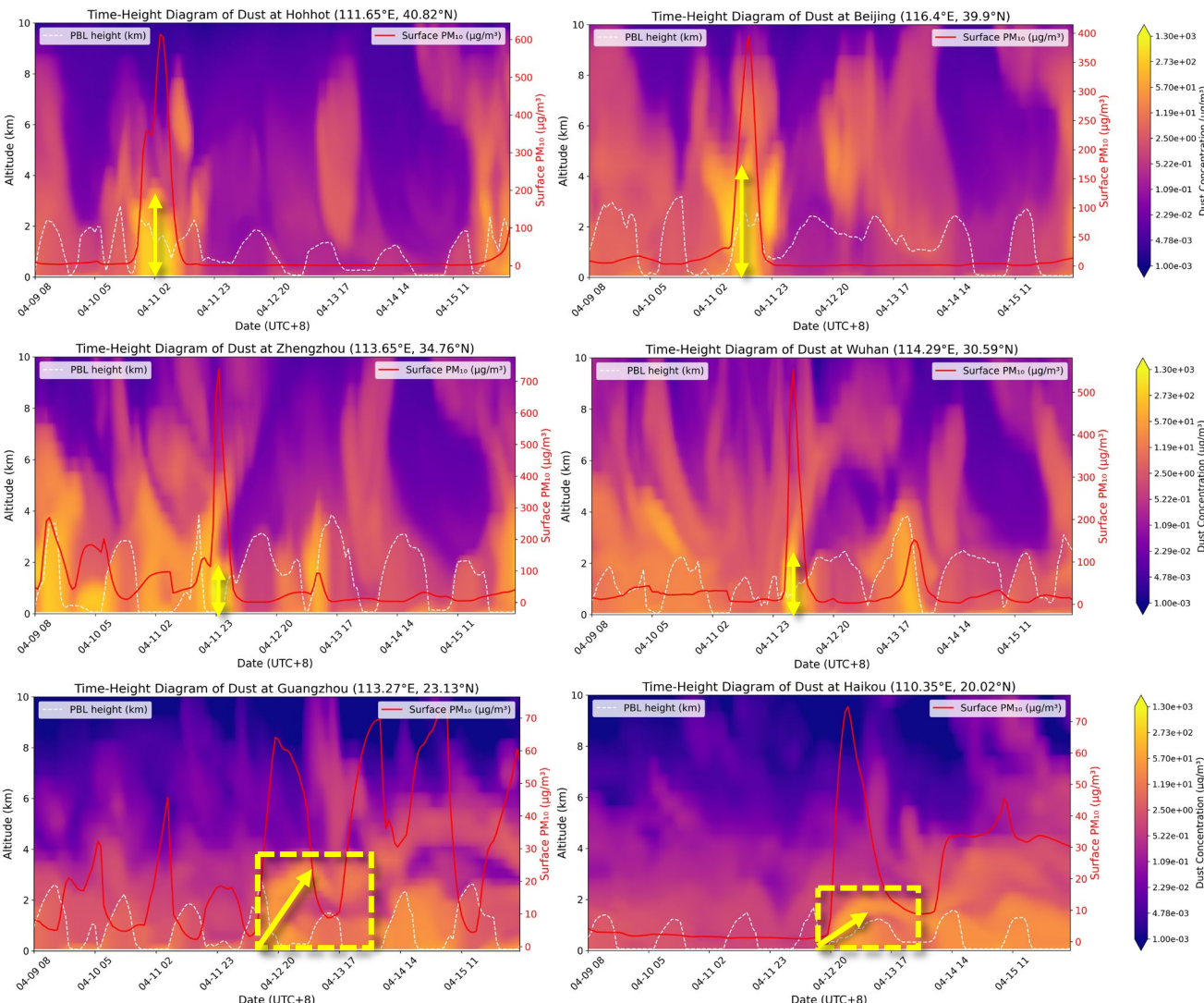
Study Regions for Vertical Evolution Analysis [Peng et al., 2026]

- ▶ **Region 1** (dust source): the starting point of vertical transport.
- ▶ **Region 2** (northern impact region): first downwind region showing downward propagation.
- ▶ **Region 3** (central impact region): evidence of upper-to-surface downward signal.
- ▶ **Region 4** (central-southern region): dust descends.
- ▶ **Region 5** (southern coastal region): longest lag, confirming progressive downward transport over distance.

April 2025 East Asia Dust Event
Accumulated Footprint & Transport Axis



Time-Height Diagrams – Evolution of Dust



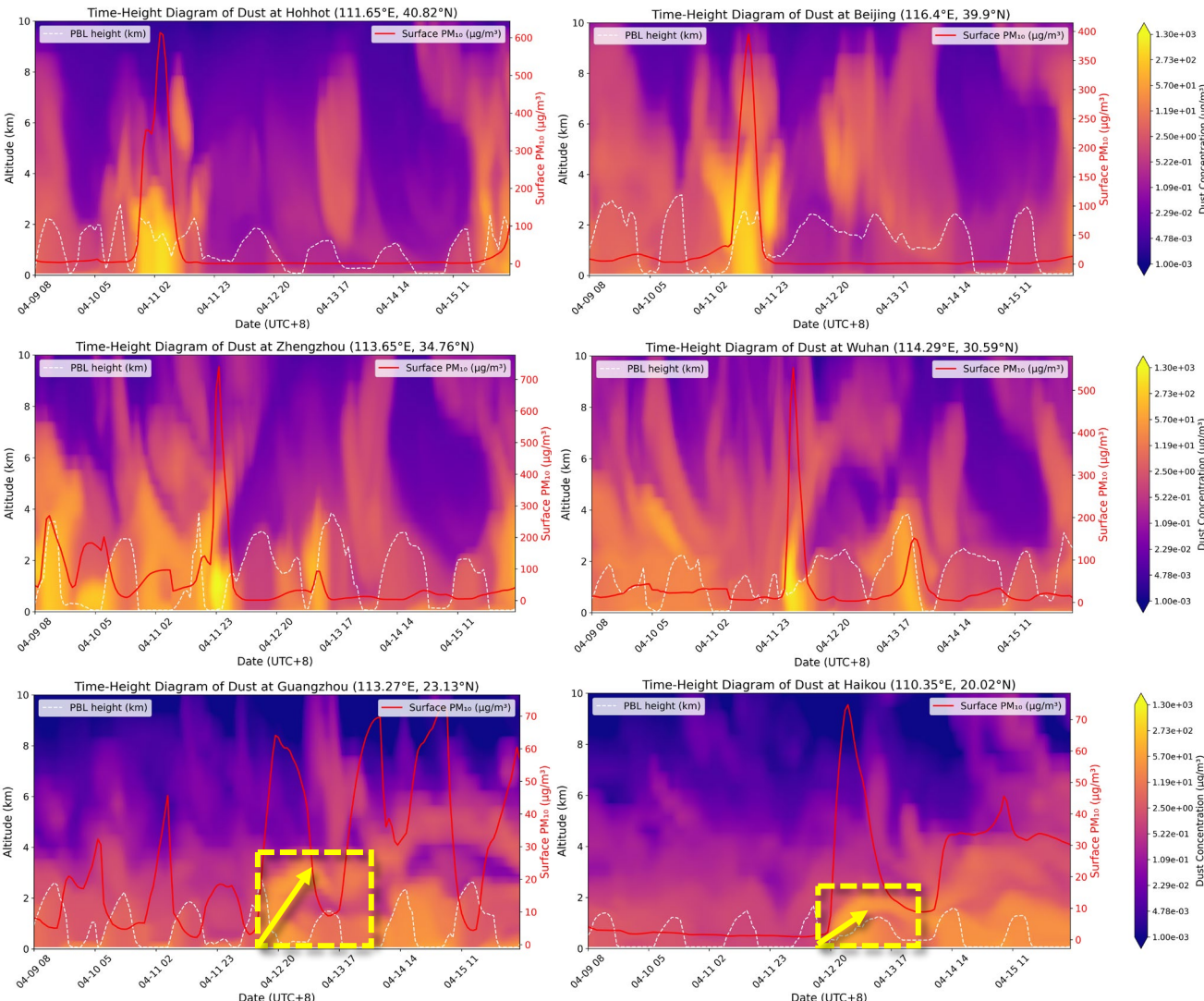
► Dust was lifted to altitudes **exceeding 3.6 km** between 35°N and 40°N on **April 11 midnight**.

► Plume reaches central Asia (around 30°N) at **April 11 late night to approximately 1.6 km**.

► Plume has **descended to 0.8 km** but remains largely in an “elevated layer”. Significantly **lower concentration** due to long range transport.

$$\text{Dust concentration} = \text{DST1} + \text{DST2} + \text{DST3} + \text{DST4}$$

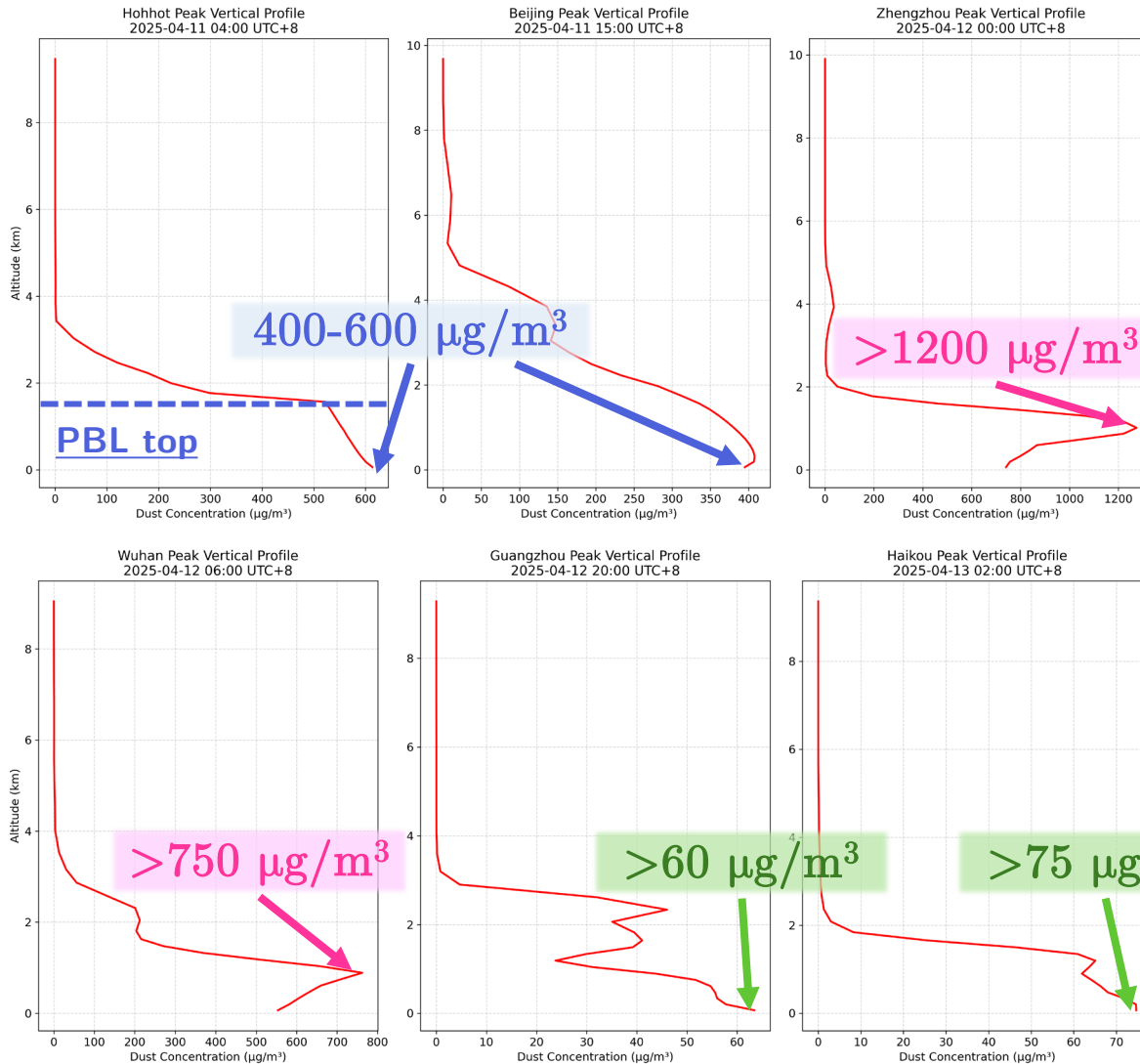
Time-Height Diagrams – Insights



- ▶ The upward transport lasts **about 12 h**. The **high-altitude transport** allows the dust to **clear high topographic barriers**.
- ▶ Highest concentration of the dust plume **detaches slightly from the surface**.
- ▶ Dust “glides over” the cooler surface air before mixing downward.

$$\text{Dust concentration} = \text{DST1} + \text{DST2} + \text{DST3} + \text{DST4}$$

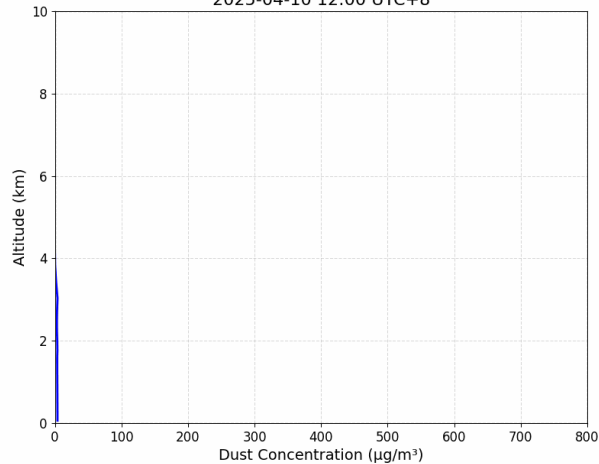
Vertical Profile of the Yellow Dragon at Peaks



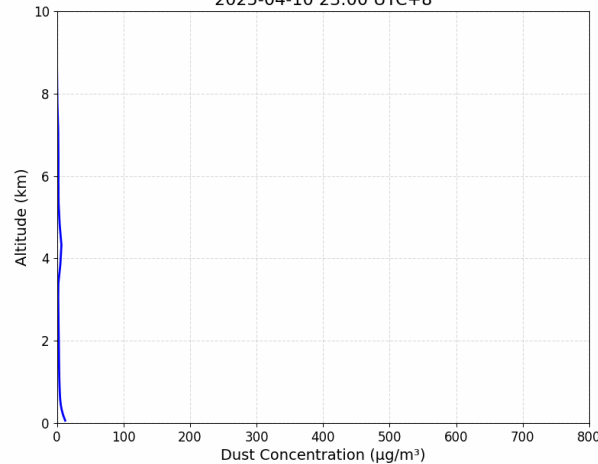
- ▶ In the source-proximal cities, the profile is nearly **well-mixed from the surface up to about 1.5 km**.
- ▶ **Sub-surface** peak in Wuhan and Zhengzhou, indicating transportation by strong lower-tropospheric wind “fumigating” dusts down to the surface.
- ▶ Guangzhou: **highest at ground, with secondary “bumps” at 1.5 km and 2.3 km. More apparent post peak concentration.**

Vertical Evolution of the Yellow Dragon

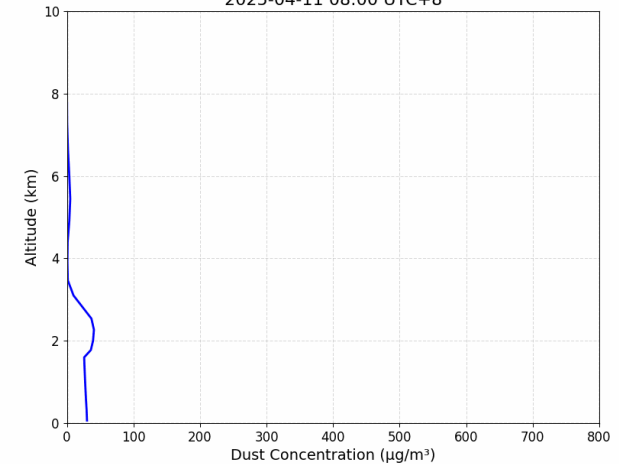
Hohhot Vertical Dust Profile
2025-04-10 12:00 UTC+8



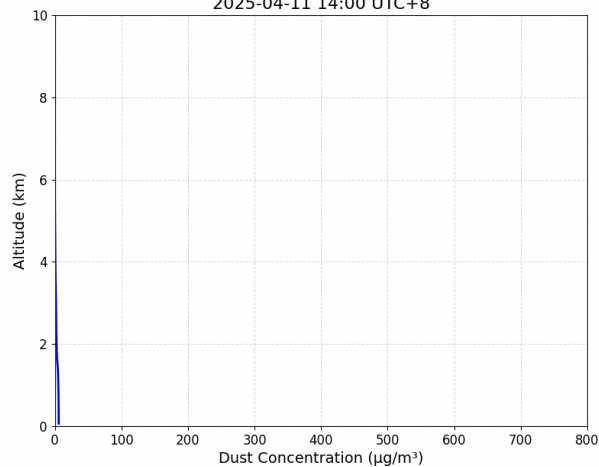
Beijing Vertical Dust Profile
2025-04-10 23:00 UTC+8



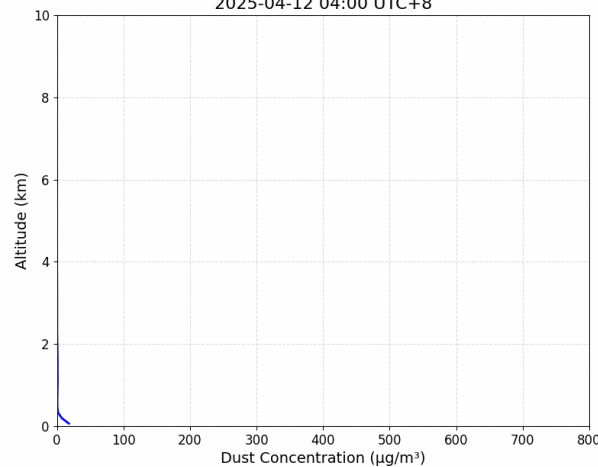
Zhengzhou Vertical Dust Profile
2025-04-11 08:00 UTC+8



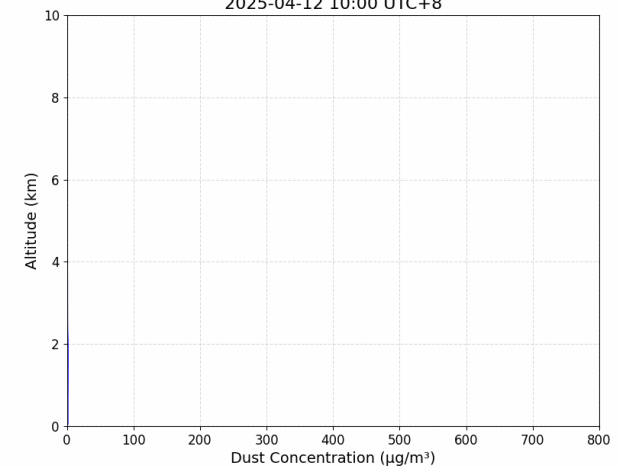
Wuhan Vertical Dust Profile
2025-04-11 14:00 UTC+8



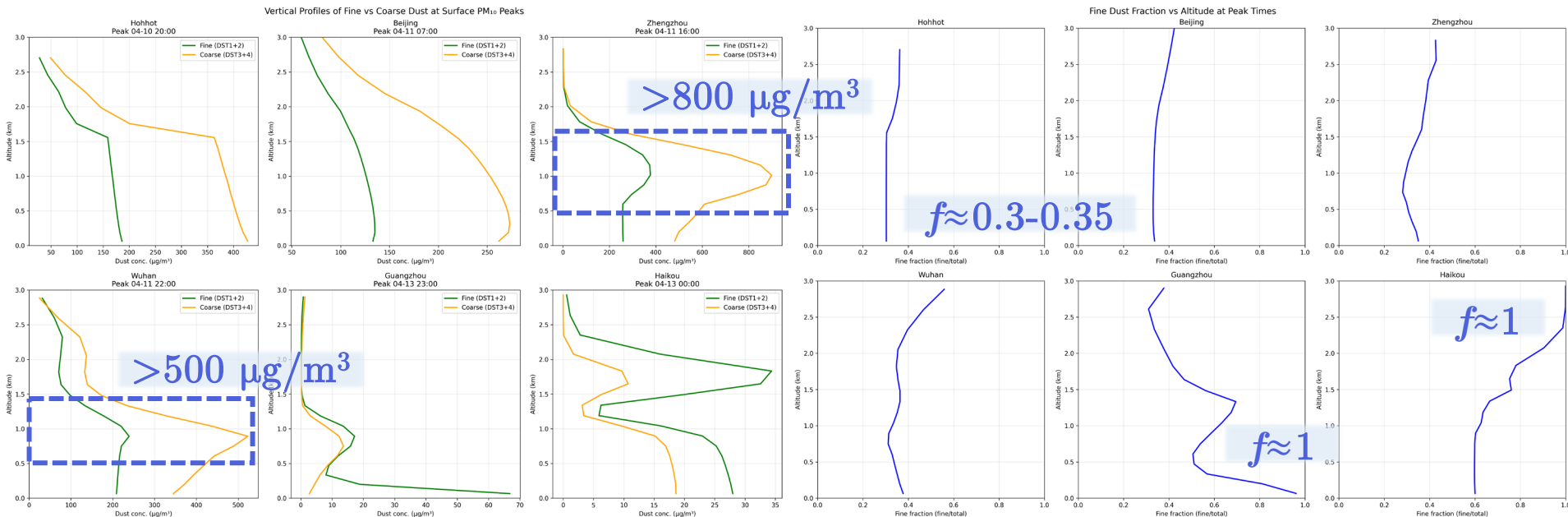
Guangzhou Vertical Dust Profile
2025-04-12 04:00 UTC+8



Haikou Vertical Dust Profile
2025-04-12 10:00 UTC+8



Size-Resolved Analysis – Gravitational Sorting

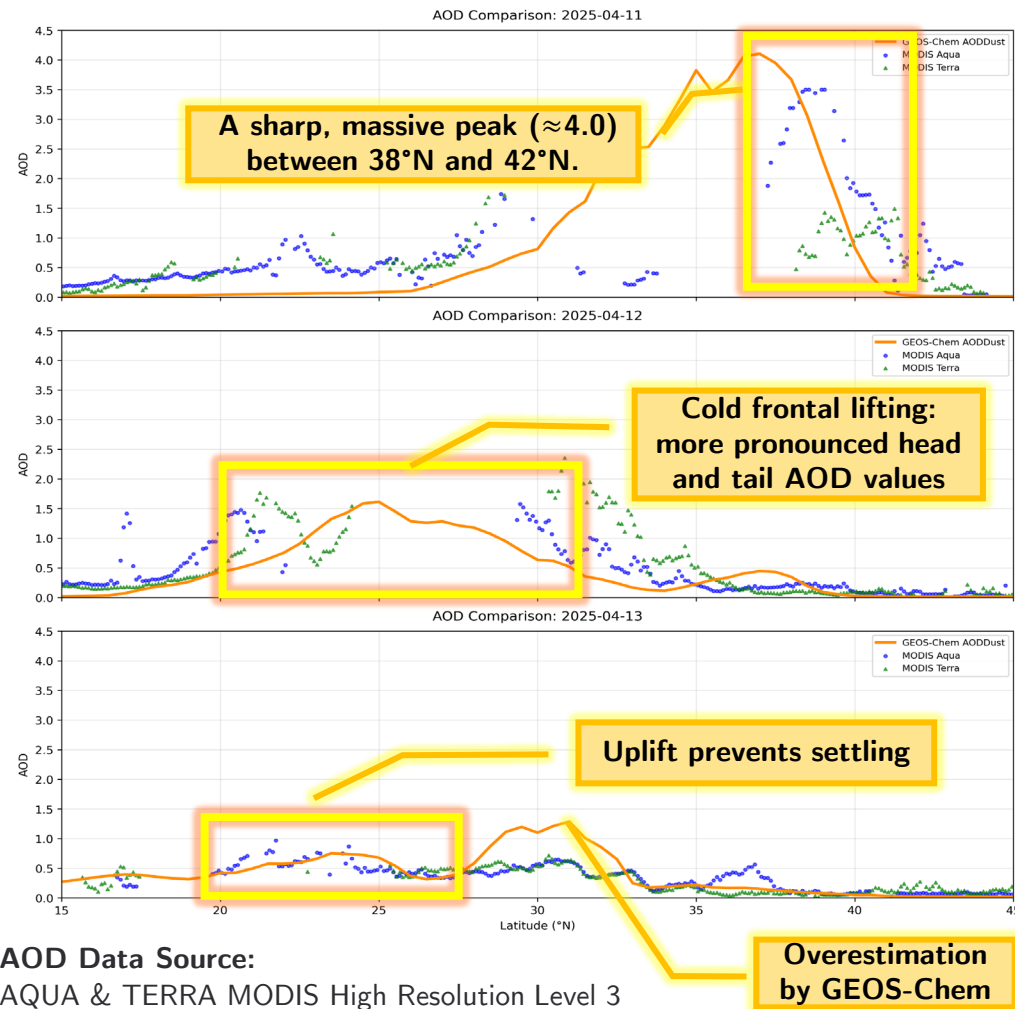
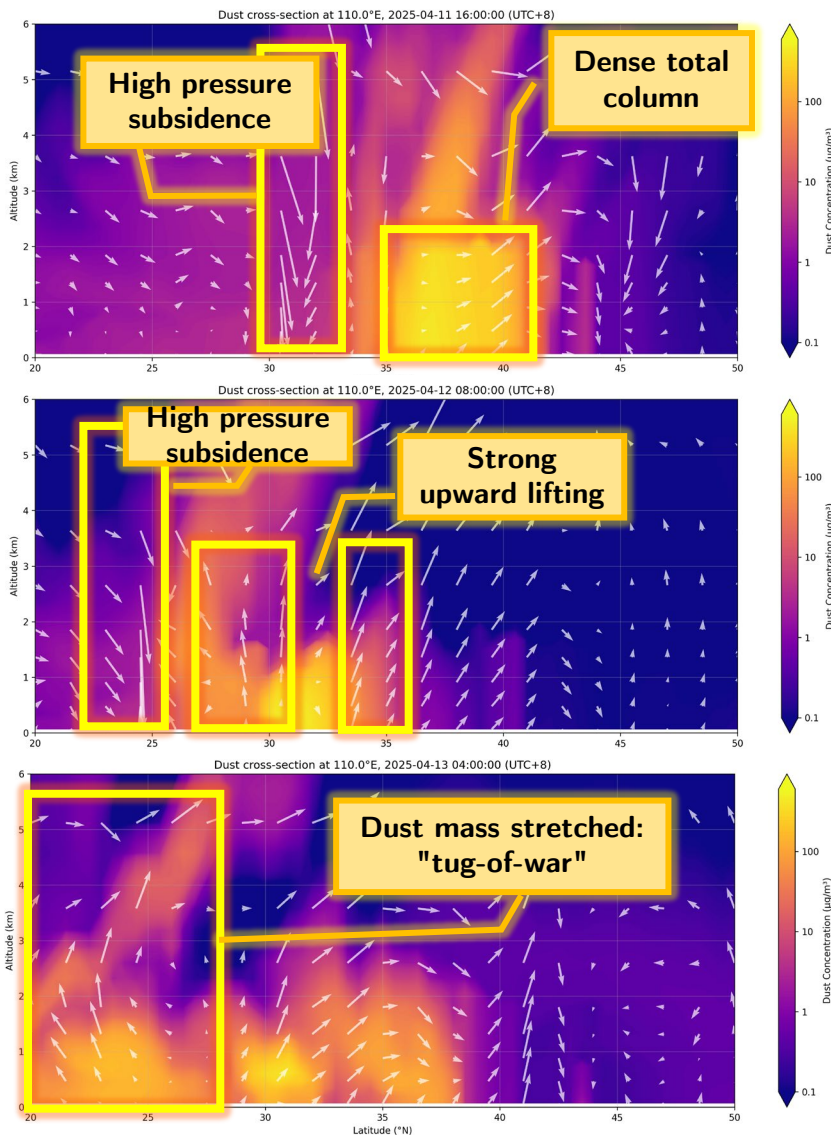


Fine Dust = DST1 + DST2, Coarse Dust = DST3 + DST4

- **Source:** “young” dust plume is dominated by coarse dust.
- **Central China:** gravitational settling, heaviest accumulation.
- **South China:** size-selective transport with wet deposition, only fine dust survives.

Insight: Gravitational settling removes coarse dust preferentially, leaving fine dust enriched downwind. Larger particles are more efficiently scavenged by the surface (impaction, interception). Coarse particles are more effective cloud condensation nuclei and are removed more efficiently in precipitation.

Latitudinal Dust Cross-Section

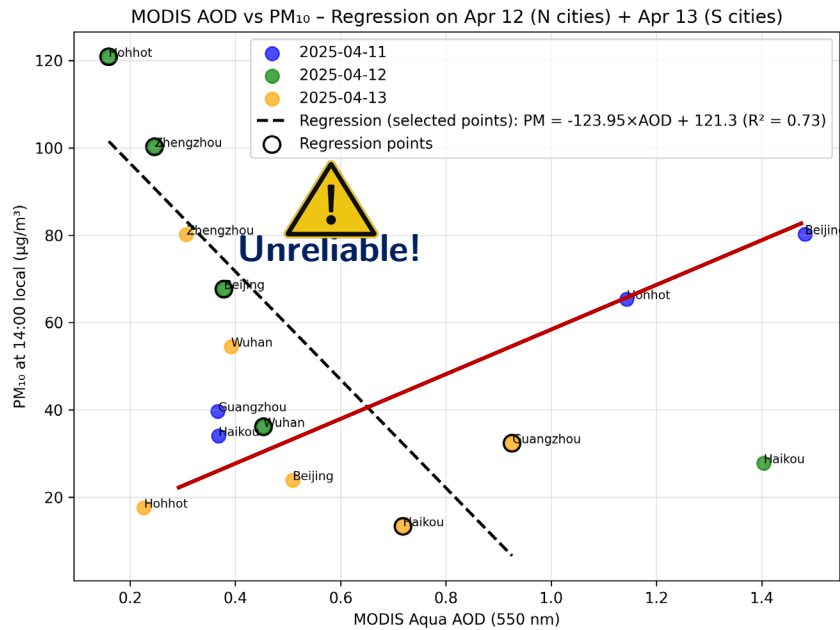


AOD Data Source:

AQUA & TERRA MODIS High Resolution Level 3 daily aerosol data, 0.1x0.1 degree grid [Gupta, 2025]

Terra passes the equator around 10:30 a.m. (local time) and Aqua around 1:30 p.m.

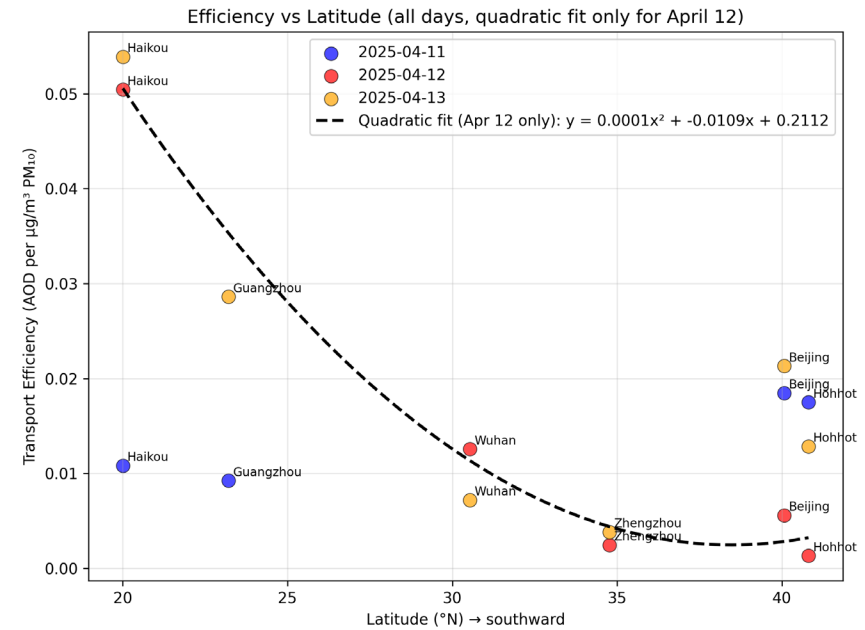
Column AOD and Surface Dust Relationship



- ▶ April 11: **AOD and PM₁₀ are both high and positively correlated** ($AOD \approx 1.5$, $PM_{10} \approx 80 \mu\text{g}/\text{m}^3$).
- ▶ April 12: High PM₁₀ ($\approx 120 \mu\text{g}/\text{m}^3$ in Hohhot) and low AOD (≈ 0.16) indicates the **dust is concentrated in a shallow, dense layer near the surface**.
- ▶ April 13: cities like Guangzhou show the opposite, high AOD (≈ 0.93) but low PM₁₀ ($\approx 32 \mu\text{g}/\text{m}^3$).

The relationship is **non-linear and stage-dependent**, **Surface-coupled** in the source/northern regions to **surface-decoupled** in the downwind/southern regions.

AOD is a poor proxy for surface health risks during dust storms.



- ▶ The “Efficiency” ratio (AOD/PM₁₀) **spikes in the south**, reaching ≈ 0.05 in Haikou compared to ≈ 0.001 in Hohhot.
- ▶ The quadratic fit for April 12 shows a **clear latitudinal gradient** where the **column load becomes increasingly disproportionate to the surface impact** as the plume moves toward 20°N.

The transport efficiency increases by an order of magnitude as the plume moves south, quantifying the degree of lofting.

Summary

- ▶ GEOS-Chem captures the event-scale source-to-downwind evolution well MODIS Aqua AOD supports the large-scale plume location and timing, although the comparison remains qualitative because AOD is a column product while the model fields are near-surface dust mass concentration.
- ▶ The time-height diagrams indicate that the dust event involved elevated transport as well as near-surface impacts, rather than simple surface advection alone. Northern cities show earlier and stronger vertical signatures, while central and southern cities respond later and more weakly, consistent with long-range transport and gradual southward extension.

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Any Questions?

Thank you!